6.1

To do

You can do this if there is a fish tank with water in it.

1 Look at the tank from the front and then very slowly move to a corner of the tank.

2 Keep looking into the tank as you do this.

3 Very gently and slowly move your head from side to side while you are standing at the corner of the tank.

4 Do this a few times and then write down what you could see.

_____

_____

_____

_____

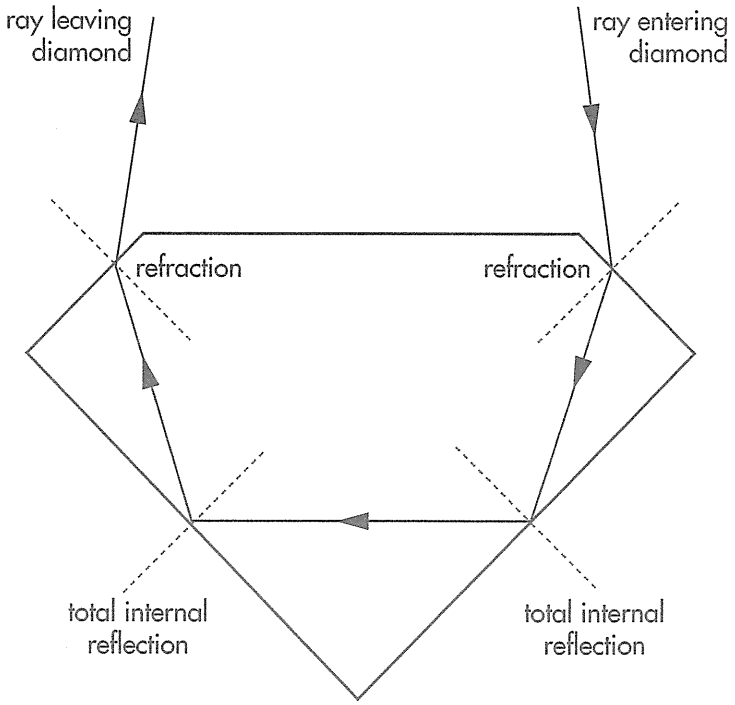
5 Where did the fish go?

Refraction can cause another tricky thing called total internal reflection to happen.

Sometimes as light passes from one substance to another, some of the light gets refracted back into the substance. When all the light gets internally refracted it is called total internal reflection. This happens in diamonds and we see it as sparkles. It is also why when you were looking at the corner of the fish tank some of the fish seemed to disappear!

What you saw was due to total internal reflection!

Diamonds sparkle because of total internal reflection



You can see from this worksheet the tricks that light can play on our eyes!

© Reed International Books Australia Pty Ltd

USING REFRACTION

6.2

In this worksheet you will learn about how refraction of light is used in everyday life.

Refraction is the bending of light as it passes from one substance to another. When you use objects that contain lenses, such as binoculars, a microscope, contact lenses or any sort of glasses, you are using refraction!

special words

refraction
converging lens
convex lens
magnified
concave lens
diverging lens

Lenses are transparent (light passes through transparent things) and have been shaped especially so that light passing through them will bend.

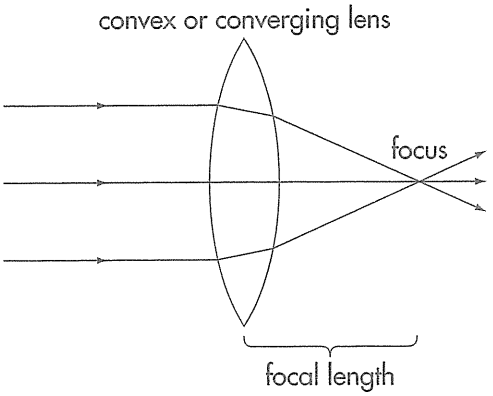
There are two different types of lenses: convex and concave.

Convex lenses

Convex lenses are called converging lenses because they make the light come together or focus at a point.

When light from something that is far away passes through a convex lens onto a screen an image can be seen. This image is upside down.

If you hold a convex lens near an object then the object is seen the *right way up* and is larger or *magnified*.



To do

Think-pair-share about convex lenses.

1 Working by yourself, write down all the uses of convex lenses that you can think of.

2 Next pair up with a partner and compare your lists.

3 Add any extra ideas that either of you had to your list.

4 Share your lists with the rest of the class.

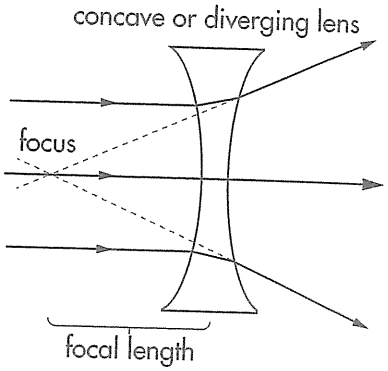
You have discovered that convex lenses are very useful. A convex lens works by making the light bend or refract inwards as it passes through the lens.

© Reed International Books Australia Pty Ltd

6.2

Concave lenses

Concave lenses are called diverging lenses. Diverging lenses make the light that passes through them spread out. Concave lenses make images that are *smaller* and the *right way up*.



To do



You will need a magnifying glass to do this activity.

1 Hold a magnifying glass up and look at objects:
   - up close
   - from a distance.

   Say what type of lens this is by looking at its shape.

2 Try it again but this time use a concave lens instead of the magnifying glass.

3 List any similarities or differences.

Concave lenses are also very useful. Concave lenses work by making the light bend or refract outwards as it passes through.

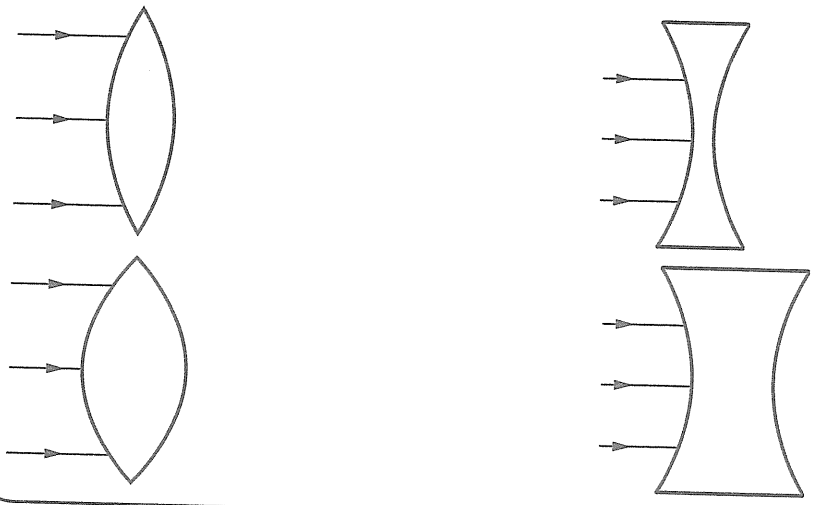


HINT: an easy way to remember the shape of the different lenses. Remember caves go into something and so concave lenses are shaped inwards.

To do



Complete the diagrams to show how the light refracts through the convex and concave lenses. Label the focus and the type of lens it is.



© Reed International Books Australia Pty Ltd

SEEING COLOURS

6.3

In this worksheet you will find out how we see colours.

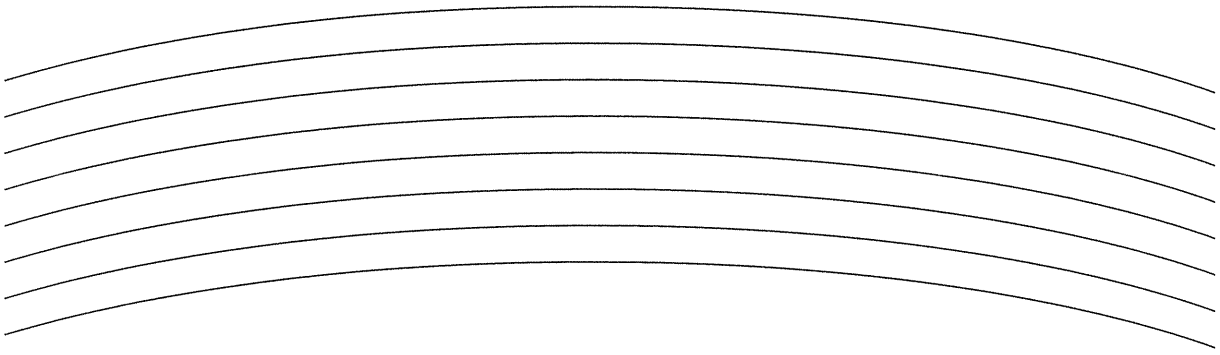
When we look at a rainbow we see many colours. There are really seven colours that we see. These are red, orange, yellow, green, blue, indigo and violet. These colours come from white light being split up or dispersed. The seven colours together are called the spectrum.

As white light passes through air, water or glass it is refracted. Different colours of light bend or are refracted different amounts and so we can see different colours.

special words

dispersed
spectrum
refracted
absorbed

Colour in the rainbow using the correct colours.



Have you noticed that sunsets are best when there are clouds or even smoke in the sky?

To do

1 Can you think of any reasons to explain the differences between sunsets? Write your explanation in the space.

_____

_____

2 Also write down any different places you have noticed a rainbow.

_____

_____

You know that white light is really made up of different coloured light. Now you will find out how you see the different colours.

In earlier years you probably learnt that light could pass through objects or be reflected by objects. Light can also go into objects and not come out. This is called being absorbed. So when light hits something it can:

- pass through the object,
- be reflected by the object or
- be absorbed by the object.

© Reed International Books Australia Pty Ltd

6.3

We only see the light that is reflected from objects. We do not see light that passes through or is absorbed by an object. If something, like a plant, looks green then green light is being reflected from the plant. The plant absorbs all the other colours of light that make up white light.

To do



1 In the space provided draw a picture of a green pear.

2 Show all the colours of white light hitting the pear like a rainbow.

3 Show that the only colour being reflected is green.

4 List all the colours that are absorbed into the pear.

5 Label your drawing carefully.

6 Complete these sentences.

- A red shirt looks red because _____

_____

_____

- Blue eyes look blue because _____

_____

_____

- We see seven colours in a rainbow because _____

_____

_____

© Reed International Books Australia Pty Ltd